

H05 Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

Complete 60% Faithfulness 2.5/5 Usefulness 2.5/5

level: hypothesis · pipeline checks 100% · updated 2026-10-04

confirmatory run on the locked holdout (2026-10-03): *INCONCLUSIVE* by the pre-registered rule. The primary pair DiD at the 03-16 split (C1) passed: separated pairs lost talk coupling relative to pairs that stayed together. The MF J_{out} criterion passed. The ...

WHAT HAPPENED

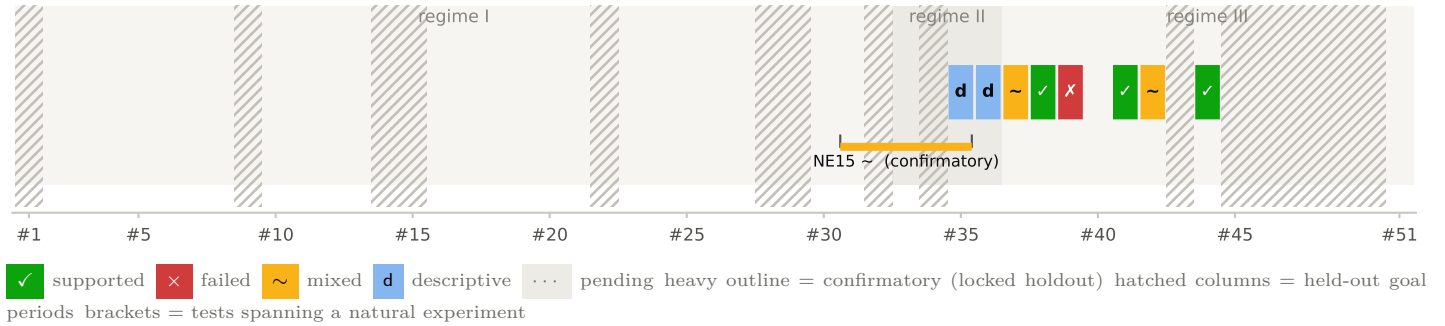
Do rooms make agents coupled blocks? Round 1 (exploratory, 99 non-holdout days, #35–#44) fitted 1-min talk and activity spins. Within-room talk coupling beat cross-room in 6/6 regime-III two-room windows (kinetic-Ising J , Fisher $p < 10^{-4}$), but per period the primary κ_x test passed only in G38, G41, G44 (mixed G37, G42; failed G39). The pooled two-way-FE effect of co-location was $\beta \approx +0.008$ ($z = 2.6$). Activity spins: not robust. Entropy production stayed at the cross-day null in 15/18 windows, so it cannot be measured at this resolution. Holdout, NE15 (03-16 #best/#rest split): the primary pair DiD passed (talk $\kappa_x -0.022$ [$-0.038, -0.008$], $p_{perm} = 0.02$) and J_{out} fell to ≈ 0 . The pooled TWFE missed ($z = 1.4$), so the pre-registered verdict is **inconclusive**. Unpredicted: within-room J_{in} rose about $\times 6$. Caveat: the post-split window (#35) was not held out and had already been seen in round 1.

PHYSICAL MODEL

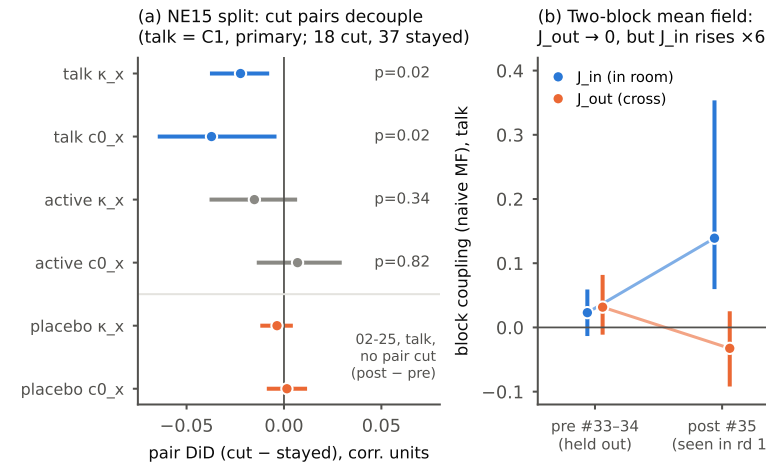
Kinetic Ising (model 02). $s_i(t) = \pm 1$: agent i talked (talk spin) or acted (activity spin) in 1-min bin t ; $s'_i = s_i(t+1)$. h_i : daily-schedule field (removed by a cross-day surrogate). r_i : room; J_{in} , J_{out} : same- and cross-room coupling; a cut predicts $J_{out} \rightarrow 0$. Σ_g : Aguilera–Ito–Kolchinsky lower bound on the entropy production σ , dual weights θ . $\hat{R}$, D : block correlations, variances. α_{ij} , γ_d : pair, day effects.

$$\begin{aligned} P(s'_i | s) &= \frac{e^{s'_i H_i}}{2 \cosh H_i}, \quad H_i = h_i + \sum_j J_{ij} s_j \\ J_{ij} &= J_{in} \delta_{r_i r_j} + J_{out} (1 - \delta_{r_i r_j}) \\ \sigma &= \sum_{i < j} (J_{ij} - J_{ji}) \langle g_{ij} \rangle, \quad g_{ij} = s'_i s_j - s_i s'_j \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle] \leq \sigma \\ J^{MF} &= D^{-1/2} (I - \hat{R}^{-1}) D^{-1/2} \\ \kappa_{ij,d} &= \alpha_{ij} + \gamma_d + \beta \text{coloc}_{ij,d} \end{aligned}$$

GOAL PERIODS TESTED 3 supported, 1 failed, 3 mixed, 2 descriptive; holdout: NE15 mixed



OBSERVABLES



WHAT THE FIGURE SHOWS

(a) Confirmatory pair difference-in-differences at the 03-16 split: the coupling change of the 18 pairs put in different rooms minus that of the 37 pairs that stayed together (pre #33–34, held out; post #35, not held out). Talk coupling dropped (C1 passed); activity did not. The 02-25 placebo, which separated no one, shows no change. Bars: day-bootstrap 95% CIs; p : assignment permutation. (b) Block mean field: cross-room J_{out} falls to ≈ 0 , while within-room J_{in} rises (unpredicted).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Candidly, agents who can't see each other's chat stop timing their chat together: close to a restatement of the scaffold, since rooms are a visibility filter. Three things are mildly useful. (1) A cheap monitor: a pair DiD on 1-min talk spins caught the cut with 18 cut pairs and 5 post-split days (power only about 25–60%). (2) A negative result an operator splitting teams should know: activity, the actual work, is *not* room-structured. Rooms reorganize conversation, not labor. (3) The unpredicted $\times 6$ rise in within-room coupling hints that attention is conserved and redirected to roommates (HH90, now tested by H18). The thermodynamic half, entropy production, is unmeasurable here and buys nothing practical.

Bottom line. Rooms cut the chat coupling, as the scaffold implies, detectably in a partly blind test; but work is not room-structured, entropy production is unmeasurable at 1-min resolution, and the confirmatory verdict was inconclusive.